

SOP-CRANE-700: Crane and Rigging Lifting Operations

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Section 1. Purpose and Scope

Governs planning and execution of crane lifts, rigging equipment inspection, and the critical-lift classification and review process.

Section 2. Lift Classification

Clause 2.1 — A lift is classified “critical” if it exceeds 75% of the crane’s rated capacity at the working radius, involves a load over a occupied structure or live process equipment, or uses two or more cranes (tandem lift).

Clause 2.2 — A critical lift requires a written lift plan reviewed by a qualified rigging engineer before the lift proceeds; a routine (non-critical) lift requires only a pre-lift check by the operator and rigger.

Section 3. Rigging Equipment Inspection

Clause 3.1 — Slings, shackles, and rigging hardware shall be visually inspected before each use and shall bear a current color-coded inspection tag no older than 12 months.

Clause 3.3 — Any sling showing cut strands, crushing, or a missing/illegible capacity tag shall be removed from service immediately, not merely flagged for the next inspection cycle.

Section 4. Exclusion Zone

Clause 4.1 — A ground-level exclusion zone equal to the maximum boom radius plus 10 feet shall be barricaded before the lift begins; personnel are prohibited from standing under a suspended load under any circumstance.

Section 5. Weather Limits

Clause 5.1 — Lifting operations shall stop when sustained wind speed exceeds the crane manufacturer’s rated limit for the load and configuration in use, or 30 mph, whichever is lower.

Section 6. Records

Clause 6.1 — Critical lift plans and rigging inspection tags are filed to the Mechanical Lifting File and retained for 3 years.